# Kaggle Competition

## Business problem

An e-commerce retailer requires analysis of its transaction data. The company is experiencing high product return rates, which directly decrease net revenue and increase reverse logistics costs. The retail industry currently faces specific structural challenges regarding online returns:

- The high cost of reverse logistics and inventory processing
- Intentional return behaviors, such as wardrobing
- The effect of aggressive promotional strategies on purchase intent and return probability
- The relationship between product quality, user reviews, and return rates
- Variability in return rates across different product categories and temporal periods

Understanding and predicting these return patterns is a requirement for the retailer to optimize profitability and operational efficiency.

## Primary task

1. Develop a **binary classification model** that accurately predicts whether a purchased item will be returned.

In addition, the business requires analysis of:

1. The key factors driving product returns. What variables most strongly indicate a high probability of a return?
2. Specific operational or marketing strategies the retailer should implement to reduce return rates, supported by your model findings and relevant retail analytics literature.

## Data

The dataset contains transactional records, customer profiles, and product information.

Key features of the data include:

- Transaction timestamps and geographic location codes
- Product categories, subcategories, pricing, and discount depths
- Customer loyalty tiers, account age, and checkout methods
- Applied promotional codes and marketing channel attribution
- Product review scores and payment methods

## Kaggle competition

Private competition with a leaderboard

Teams with up to 3 members (but you can also form a solo person team)

How does Kaggle work

- 10 submissions allowed per day to the **public** leaderboard
- 2 final models considered for **private** leaderboard for final evaluation

Deadline for final Kaggle submission, Friday, March 20th at noon

## Grading

This final project is worth 30 points (30% of the overall grade)

- Executive summary report (15 points)
- Model performance based on AUC (15 points)
  - Detailed score will be announced soon

Up to 5 extra credits for the top 5 teams

- 1 credit for the Top 5, up to 5 credits for the winning team

## Executive summary report (15 points)

Build a slide deck (max 5 slides + title) covering the following topics:

- Data understanding (visualization, etc.)
- Data preparation (variable treatment, feature creation)
- Modeling (what model and algorithm?)
- Evaluation methodology (tuning, benchmark)
- Managerial implication and suggestions

Imagine you are presenting to executive board and need to summarize on how you derived your results and what the management should do about. Consider your slides to be meant as a handout (more dense information).

Upload to BlackBoard by Friday, March 20th at midnight

## Model performance (15 points)

**Model Performance Grading Scale (15 points)**

Points are obtained based on the performance of AUC (Area Under the ROC Curve).

| AUC | Awarded points |
|---|---|
| AUC < 0.59 | 0 |
| 0.59 ≤ AUC < 0.63 | 2 |
| 0.63 ≤ AUC < 0.65 | 3 |
| 0.65 ≤ AUC < 0.68 | 4 |
| 0.68 ≤ AUC < 0.70 | 5 |
| 0.70 ≤ AUC < 0.72 | 6 |
| 0.72 ≤ AUC < 0.74 | 7 |
| 0.74 ≤ AUC < 0.75 | 8 |
| 0.75 ≤ AUC < 0.77 | 9 |
| 0.77 ≤ AUC < 0.78 | 10 |
| 0.78 ≤ AUC < 0.80 | 11 |
| 0.80 ≤ AUC < 0.81 | 12 |
| 0.81 ≤ AUC < 0.82 | 13 |
| 0.82 ≤ AUC < 0.84 | 14 |
| AUC ≥ 0.84 | 15 |

Baseline model performance ~0.67 AUC

- Improve through feature engineering and model optimization.

Advanced models' performance ~0.84 AUC

If nobody reaches an AUC of 0.84, I will compress the scale to the best performing model (i.e., at least one team will have 15pts)

## How to access it

Create a Kaggle account (www.kaggle.com)

- Access the competition using the link on Blackboard
- Add your team members and set your team name (3 people max.)

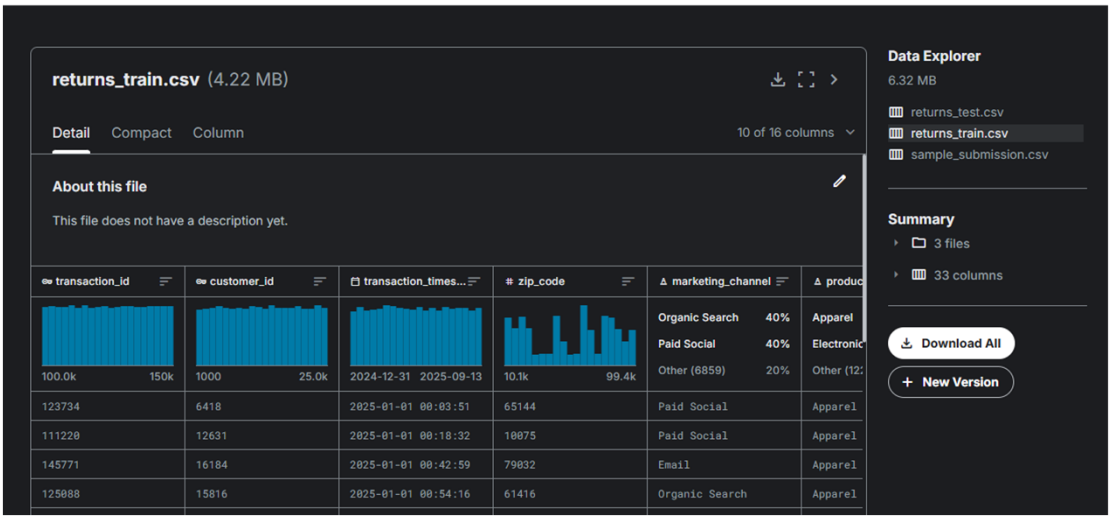
3 Data files are provided

- returns_train.csv → This is your training dataset to build your model
- returns_test.csv → Unseen data where you predict the target variable
- sample_submission.csv → Formatting sample for your prediction submission

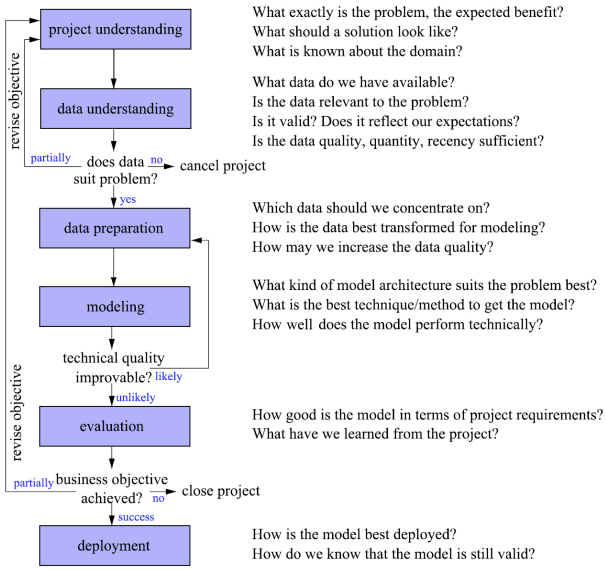
To make a submission, you need to upload a CSV:

- Two columns: **id** and **returned** (see sample submission)
- IMPORTANT: AUC requires probabilities

## Files available



# CRoss Industry Standard Process for Data Mining




Source: Berthold et al. (2010) Guide to Intelligent Data Analysis

# Some suggestions

Don't wait until the very end to submit models

- submissions per day are limited – and they count per team
- It is an iterative process → don't let yourself be discouraged by first submission
- Split tasks amongst your group

Understanding the dataset is key!

- Inspect dataset consistencies and data types (timestamps, ordinal etc.)
- Think carefully about how you encode variables
- Be smart in treating missing values
- Timestamps can contain valuable information
- Class imbalance

## Some suggestions cont.

Importance of the evaluation setup

- Choice of evaluation metric
- Model selection choice

Report writing:

- Should be of academic rigor
- Allow that someone else would be able to build the same model without code
  - Understand what went into your model
  - How did you treat your data
  - How did you evaluate it
- Managerial implications
  - Think beyond just predictive accuracy (what should management do)
  - Next steps (additional analysis)
  - Search academic literature to support your managerial implications